**Project Report: AI Laptop Price Predictor**

**Course:** CSA2001 - Fundamentals in AI and ML

**Student Name:** Alok Raj

**Registration Number:** 25BCE11089

**Institution:** VIT Bhopal University

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## 1. Problem Statement

In the contemporary consumer electronics market, laptop pricing often lacks transparency, making it difficult for users to assess if a device's price is justified by its internal hardware. This project aims to build a **Supervised Learning** agent that predicts the market price of a laptop based on three critical features: **RAM (GB)**, **CPU Speed (GHz)**, and **SSD Capacity (GB)**.

The goal is to provide a data-driven estimation tool that categorizes laptops into **Budget**, **Mid-range**, or **High-end** tiers, helping consumers make informed **Statistical Decisions**.

## 2. Why it Matters

As a **B.Tech CSE Core student**, understanding the relationship between hardware specifications and market value is a fundamental engineering challenge. This project:

- Applies **Linear Algebra and Probability** to a real-world retail problem.
- Demonstrates how **Machine Learning** can automate complex valuation tasks.
- Provides transparency to consumers, reducing the gap between hardware cost and retail pricing.

## 3. Technical Approach & Methodology

The project follows the standard **Machine Learning pipeline** outlined in the course syllabus:

- **Data Representation:** Hardware specs are treated as independent features ($X$) and the price as the target variable ($y$).
- **Algorithm Selection:** I implemented **Linear Regression**, a supervised learning technique suitable for predicting continuous numerical values like price.

- **Model Training:** The dataset was split using train_test_split (80% training, 20% testing) to ensure the model generalizes well to new data.
- **Evaluation Metrics:** The model's performance is measured using **Mean Absolute Error (MAE)** and the **R-squared ($R^2$) Score**.

**4. Key Decisions & Features**

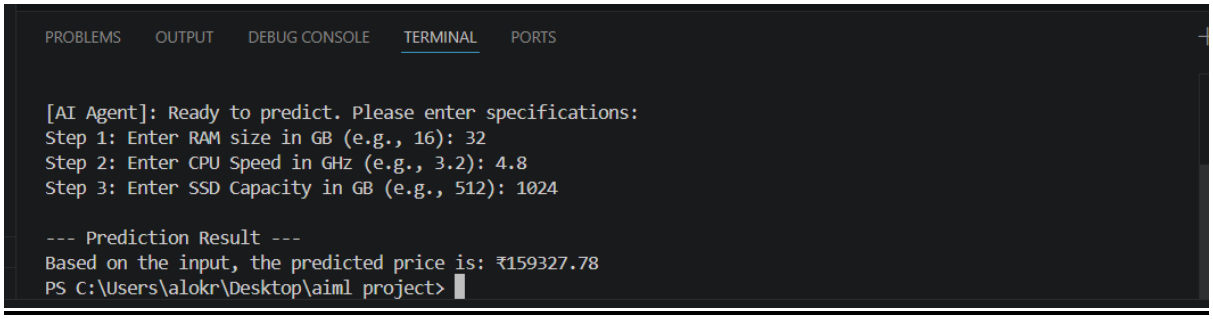
- **Feature Importance:** The script calculates and displays coefficients to show which hardware component (e.g., RAM vs. SSD) has the highest impact on the price.
- **Robust Input Handling:** A get_input function with a try-except block ensures the agent only processes valid numerical data, preventing runtime crashes.
- **Categorical Logic:** Based on the predicted price, the system automatically classifies the device into a market segment (Budget/Mid/High-end).

**5. Challenges Faced**

- **Overfitting vs. Underfitting:** Balancing the training data to ensure the model predicts accurately for both low-end and high-end laptops.
- **Data Normalization:** Ensuring that different units (GB for RAM vs. GHz for CPU) were handled correctly by the **Linear Regression** estimator.

**6. Learning Outcomes**

Through this project, I have achieved the following **Course Outcomes (COs)**:

- **CO4:** Designed a problem-solving agent using AI techniques.
- **CO5:** Applied Machine Learning algorithms to solve a real-world engineering problem.
- **Practical Skills:** Gained proficiency in **Python**, **Pandas**, and **Scikit**

**<u>Code execution screenshot :</u>**